

Group Lab Report

EXPERIMENT 10

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AIM-Set up Docker container in the local machine and pulling various images from Docker Hub using internet and login into newly created images.

REQUIREMENTS-

- **OS:** Windows 10/11, macOS, or Linux.
- **Hardware:** Minimum 4GB RAM (8GB recommended).
- **Internet:** Required to pull images from Docker Hub.
- **BIOS Setting:** Ensure "Virtualization Technology" (VT-x/AMD-V) is enabled in your laptop's BIOS settings.

IMPLEMENTATION-

- **Installation:** Download and install **Docker Desktop**.
- **Verification:** Run **docker --version** to ensure it's alive.
- **Pulling Images:** Use the CLI to fetch tiny, pre-configured operating systems:
 1. **docker pull alpine**
 2. **docker pull ubuntu**
- **Running/Login:** ***docker run -it ubuntu**, This command instantly drops you into a live terminal inside that "machine."

SCREENSHOTS-

```
C:\Users\laali>docker --version
Docker version 29.3.1, build c2be9cc
```

```
C:\Users\laali>docker search ubuntu
```

NAME	DESCRIPTION	STARS	OFFICIAL
ubuntu	Ubuntu is a Debian-based Linux operating sys...	17800	[OK]
ubuntu/squid	Squid is a caching proxy for the Web. Long-t...	124	
ubuntu/nginx	Nginx, a high-performance reverse proxy & we...	141	
ubuntu/bind9	BIND 9 is a very flexible, full-featured DNS...	117	
ubuntu/kafka	Apache Kafka, a distributed event streaming ...	60	
ubuntu/apache2	Apache, a secure & extensible open-source HT...	102	
ubuntu/prometheus	Prometheus is a systems and service monitori...	78	
ubuntu/zookeeper	ZooKeeper maintains configuration informatio...	14	
ubuntu/jre	Chiseled Java runtime based on Ubuntu. Long...	25	
ubuntu/dotnet-aspnet	Chiselled Ubuntu runtime image for ASP.NET a...	26	
ubuntu/mysql	MySQL open source fast, stable, multi-thread...	73	
ubuntu/postgres	PostgreSQL is an open source object-relatio...	43	
ubuntu/python	A chiselled Ubuntu rock with the Python runt...	31	
ubuntu/redis	Redis, an open source key-value store. Long...	25	
ubuntu/dotnet-runtime	Chiselled Ubuntu runtime image for .NET apps...	24	
ubuntu/dotnet-deps	Chiselled Ubuntu for self-contained .NET & A...	16	
ubuntu/grafana	Grafana, a feature rich metrics dashboard & ...	12	
ubuntu/memcached	Memcached, in-memory keyvalue store for smal...	6	
ubuntu/loki	Grafana Loki, a log aggregation system like ...	2	
ubuntu/mimir	Ubuntu ROCK for Mimir, a horizontally scalab...	0	
ubuntu/grafana-agent	Ubuntu ROCK for Grafana Agent, an open-sourc...	0	
ubuntu/alertmanager	Ubuntu Rock for Alertmanager. It handles ale...	0	
ubuntu/jdk	Ubuntu rock for OpenJDK	0	
ubuntu/opentelemetry-collector	This is a rock for the OpenTelemetry Collect...	0	
ubuntu/node	Ubuntu-based Node.js image for server-side a...	4	

```
C:\Users\laali>docker pull ubuntu
```

```
Using default tag: latest
```

```
latest: Pulling from library/ubuntu
```

```
817807f3c64e: Pull complete
```

```
e4a1e8de092c: Download complete
```

```
Digest: sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c
```

```
Status: Downloaded newer image for ubuntu:latest
```

```
docker.io/library/ubuntu:latest
```

```
C:\Users\laali>docker pull alpine
Using default tag: latest
latest: Pulling from library/alpine
589002ba0eae: Pull complete
caa817ad3aea: Download complete
9e595aac14e0: Download complete
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
docker.io/library/alpine:latest
```

```
C:\Users\laali>docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:452a468a4bf985040037cb6d5392410206e47db9bf5b7278d281f94d1c2d0931
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
```

```
C:\Users\laali>docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	25109184c71b	13.1MB	3.95MB	
hello-world:latest	452a468a4bf9	25.9kB	9.49kB	
ubuntu:latest	186072bba1b2	119MB	31.7MB	

 Info →  In Use

```

C:\Users\laali>docker run -it ubuntu /bin/bash
root@744b0e12cf5d:/# -i
bash: -i: command not found
root@744b0e12cf5d:/# i
bash: i: command not found
root@744b0e12cf5d:/# t
bash: t: command not found
root@744b0e12cf5d:/# ls -l
total 48
lrwxrwxrwx    1 root root      7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x    5 root root 4096 Apr 22  2024 boot
drwxr-xr-x    2 root root  360 Mar 30 15:37 dev
drwxr-xr-x    1 root root 4096 Mar 30 15:37 etc
drwxr-xr-x    3 root root 4096 Feb 17 02:09 home
lrwxrwxrwx    1 root root      7 Apr 22  2024 lib -> usr/lib
lrwxrwxrwx    1 root root      9 Apr 22  2024 lib64 -> usr/lib64
drwxr-xr-x    2 root root 4096 Feb 17 02:02 media
drwxr-xr-x    2 root root 4096 Feb 17 02:02 mnt
drwxr-xr-x    2 root root 4096 Feb 17 02:02 opt
dr-xr-xr-x 296 root root      0 Mar 30 15:37 proc
drwx-----   2 root root 4096 Feb 17 02:09 root
drwxr-xr-x    4 root root 4096 Feb 17 02:09 run
lrwxrwxrwx    1 root root      8 Apr 22  2024 sbin -> usr/sbin
drwxr-xr-x    2 root root 4096 Feb 17 02:02 srv
dr-xr-xr-x   13 root root      0 Mar 30 15:37 sys
drwxrwxrwt    2 root root 4096 Feb 17 02:09 tmp
drwxr-xr-x   12 root root 4096 Feb 17 02:02 usr
drwxr-xr-x   11 root root 4096 Feb 17 02:09 var
root@744b0e12cf5d:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
root@744b0e12cf5d:/# docker run -it alpine sh
bash: docker: command not found
root@744b0e12cf5d:/# exit
exit

```

```

C:\Users\laali>docker run -it alpine sh
/ # ls
bin      etc      lib      mnt      proc     run      srv      tmp      var
dev      home    media    opt      root     sbin     sys      usr
/ # exit

```

```
C:\Users\laali>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ecd16db8f030	hello-world	"sh"	25 seconds ago	Created		awesome_hugle
72cd9bc8c90c	alpine	"sh"	53 seconds ago	Exited (0) 41 seconds ago		busy_moore
744b0e12cf5d	ubuntu	"/bin/bash"	2 minutes ago	Exited (127) About a minute ago		loving_ardinghelli

```
C:\Users\laali>docker stop --help
Usage:  docker stop [OPTIONS] CONTAINER [CONTAINER...]

Stop one or more running containers

Aliases:
  docker container stop, docker stop

Options:
  -s, --signal string  Signal to send to the container
  -t, --timeout int    Seconds to wait before killing the container
```

```
C:\Users\laali>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ecd16db8f030	hello-world	"sh"	8 minutes ago	Created		awesome_hugle
72cd9bc8c90c	alpine	"sh"	8 minutes ago	Exited (0) 8 minutes ago		busy_moore
744b0e12cf5d	ubuntu	"/bin/bash"	10 minutes ago	Exited (127) 9 minutes ago		loving_ardinghelli

```
C:\Users\laali>docker start 72c
72c
C:\Users\laali>docker attach 72c
/ # exit
C:\Users\laali>docker rm 72c 744
72c
744
```

```
C:\Users\laali>docker rmi ubuntu alpine hello-world
Untagged: ubuntu:latest
Deleted: sha256:186072bba1b2f436cbb91ef2567abca677337cf786c86e107d25b7072feef0c
Untagged: alpine:latest
Deleted: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Error response from daemon: conflict: unable to delete hello-world:latest (must be forced) - container ecd16db8f030 is using its referenced image 452a468a4bf9

C:\Users\laali>docker rm ecd
ecd

C:\Users\laali>docker rmi ubuntu alpine hello-world
Untagged: hello-world:latest
Deleted: sha256:452a468a4bf985040037cb6d5392410206e47db9bf5b7278d281f94d1c2d0931
Error response from daemon: No such image: ubuntu:latest
Error response from daemon: No such image: alpine:latest
```

```
C:\Users\laali>docker system prune -a
WARNING! This will remove:
- all stopped containers
- all networks not used by at least one container
- all images without at least one container associated to them
- all build cache

Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B
```

RESULT- The experiment to set up Docker container in the local machine and pulling various images from Docker Hub using internet and login into newly created images was successfully implemented and output was obtained.